Hybrid Nonlinear Conjugate Gradient and Negative Curvature Descent for Strict Saddle Functions: Complexity Analysis

For Clément and I to see where we are, and where we are going

In this work, we revisit the Hybrid Gradient and Negative Curvature Descent from [3] using the Non-linear Conjugate Gradient with modified restart from [1]. The first part is an analysis of the complexity of the Hybrid Non-Linear Conjugate Gradient (NCG) and Negative Curvature Descent (NCD) algorithm for an arbitrary non-convex function. We study this setting under the hypothesis of known Lipschitz constants. Therefore, no line search is performed in this case. This case is trivial as the complexity of the algorithm is simply the sum of the complexities found in the two papers mentioned:

$$K = O(\epsilon_g^{-2} + \epsilon_g^{-2(1+p-q)} + \epsilon_H^{-3}) \tag{1}$$

Subsequently, we analyze the hybrid framework under the hypothesis that the non-convex function follows the strict saddle property. This setting gives us three regimes to analyze based on the strit saddle conditions:

- **Regime 1**: High gradient. In this case, we apply the standard NCG algorithm considered in [1]. We get the same complexity (up to constants):

$$K = O(\alpha^{-2} + \alpha^{-2(1+p-q)}) \tag{2}$$

- **Regime 2**: Negative Curvature. In this case, we apply the standard NCD algorithm considered in [3]. We get the same complexity:

$$K = O(\beta^{-3}) \tag{3}$$

- **Regime 3**: Local strong convexity. The complexity of this regime is partitioned into 3 cases based on values of p and q:
  - The power $1 + p - q = 1$: This case gives us the same rate as the restarted case (i.e. Gradient Descent)
  - The power $1 + p - q$ is different from one. This gives us new complexities shown in Section 5.

In Section 6, we first prove that once we enter regime 3, we stay in the strong convexity region $B(x^\star, 2\delta)$. We also show that it is not possible to prove that once we enter Regime 3, we stay in regime 3 from the first iteration done in this regime ($B(x^\star, \delta)$). However, we find specific values for the number of iterations done in regime 3 after which we stay in regime 3 for both the restarted and non-restarted cases.

1 Setting and Assumptions

We consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \tag{4}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable. We write $g_k = \nabla f(x_k)$ for the gradient at iterate x_k, $x^\star$ the minimizer of f and $f(x^\star)$ the minimum value of f.

We make the following standing assumptions throughout.

A1. ∇f is L_1-Lipschitz continuous: $\|\nabla f(x) - \nabla f(y)\| \leq L_1\|x - y\|$ for all $x, y \in \mathbb{R}^n$.

A2. $\nabla^2 f$ is L_2-Lipschitz continuous: $\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L_2\|x - y\|$ for all $x, y \in \mathbb{R}^n$.

A3. f is bounded below: there exists $f(x^\star) \in \mathbb{R}$ such that $f(x) \geq f(x^\star)$ for all $x \in \mathbb{R}^n$.

We denote the gap to the minimum by $\Delta_i := f(x_i) - f(x^\star)$.

1.1 Partition of iterations

Following [1], we partition the iterations into two disjoint sets based on the restart condition. Given parameters $\sigma \in (0, 1]$, $\kappa \geq 1$, $p \geq 0$, $q \geq 0$, we define:

$$\mathcal{N} = \{k \in \mathbb{N} \mid g_k^T d_k \leq -\sigma\|g_k\|^{1+p} \text{ and } \|d_k\| \leq \kappa\|g_k\|^q\}, \qquad \mathcal{R} = \mathbb{N} \setminus \mathcal{N}. \tag{5}$$

We call $k \in \mathcal{N}$ a *non-restarted* iteration and $k \in \mathcal{R}$ a *restarted* iteration (where d_k is reset to $-g_k$).

1.2 Main inequalities

The following inequalities are used repeatedly. From A1 (descent lemma), for any x and direction d:

$$f(x + \alpha d) \leq f(x) + \alpha \nabla f(x)^T d + \frac{L_1}{2}\alpha^2\|d\|^2. \tag{6}$$

From L_1-smoothness, for any x in a neighborhood of a local minimizer $x^\star$:

$$\|\nabla f(x)\|^2 \leq 2L_1(f(x) - f(x^\star)). \tag{7}$$

If additionally f is γ-strongly convex in that neighborhood (Polyak–Łojasiewicz inequality):

$$\|\nabla f(x)\|^2 \geq 2\gamma(f(x) - f(x^\star)). \tag{8}$$

2 Algorithm

The restart condition is:

$$g_{k+1}^T d_{k+1} \geq -\sigma\|g_{k+1}\|^{1+p} \quad \text{or} \quad \|d_{k+1}\| \geq \kappa\|g_{k+1}\|^q. \tag{9}$$

Algorithm 1: Hybrid NCG + Negative Curvature Descent

Input: $x_0 \in \mathbb{R}^n$, $\sigma \in (0,1]$, $\kappa \ge 1$, $p \ge 0$, $q \ge 0$, $\varepsilon_g > 0$, $\varepsilon_H > 0$.

Set $g_0 = \nabla f(x_0)$, $d_0 = -g_0$.

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for k = 0, 1, 2, ... do
    Compute g_k = ∇f(x_k).
    if ‖g_k‖ ≥ ε_g then
        /* NCG phase                                              */
        Compute stepsize α_k > 0 (via line search or optimal fixed step).
        Set x_{k+1} = x_k + α_k d_k and g_{k+1} = ∇f(x_{k+1}).
        Choose conjugate direction parameter β_{k+1}.
        Set d_{k+1} = −g_{k+1} + β_{k+1} d_k.
        if condition (9) holds then
            Restart: d_{k+1} ← −g_{k+1}.
    else
        Compute smallest eigenvalue λ_k and eigenvector e_k of ∇²f(x_k), with sign chosen such that
          ⟨g_k, e_k⟩ ≤ 0.
        if −λ_k ≥ ε_H then
            /* Negative curvature phase                           */
            x_{k+1} = x_k + (2λ_k / L_2) e_k.
            Reset d_{k+1} = −∇f(x_{k+1}).
        else
            Stop and return x_⋆ = x_k.
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Convergence guarantee: $\|\nabla f(x_\star)\| \le \varepsilon_g$, $-\lambda_{\min}(\nabla^2 f(x_\star)) \le \varepsilon_H$.

Stepsize. Without line search, using the descent lemma (6) and optimizing over α, the optimal fixed stepsize for non-restarted iterations is:

$$\alpha^\star_{\mathcal{N},k} = \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q} \tag{10}$$

while for restarted iterations ($d_k = -g_k$), the stepsize is $\alpha^\star_{\mathcal{R},k} = 1/L_1$. Both choices require knowledge of L_1 but not a line search. The complexity constants are then:

$$c_{\mathcal{R}} = \frac{1}{2L_1}, \qquad c_{\mathcal{N}} = \frac{\sigma^2}{2L_1\kappa^2}. \tag{11}$$

3 Per-Iteration Decrease Guarantees

We establish decrease bounds for each type of iteration.

3.1 NCG phase: non-restarted iterations ($k \in \mathcal{N}$)

Lemma 1. *Let A1 hold and $k \in \mathcal{N}$ with $\|g_k\| > 0$. Then:*

$$f(x_k) - f(x_{k+1}) \ge c_{\mathcal{N}}\,\|g_k\|^{2(1+p-q)}. \tag{12}$$

Proof. From the descent lemma (6), using $g_k^T d_k \le -\sigma\|g_k\|^{1+p}$ and $\|d_k\| \le \kappa\|g_k\|^q$:

$$f(x_{k+1}) \le f(x_k) - \alpha\sigma\|g_k\|^{1+p} + \frac{L_1}{2}\alpha^2\kappa^2\|g_k\|^{2q}.$$

Setting the derivative with respect to α to zero gives $\alpha^\star = \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q}$. Substituting:

$$f(x_{k+1}) \le f(x_k) - \frac{\sigma^2}{L_1\kappa^2}\|g_k\|^{2(1+p-q)} + \frac{\sigma^2}{2L_1\kappa^2}\|g_k\|^{2(1+p-q)} = f(x_k) - \frac{\sigma^2}{2L_1\kappa^2}\|g_k\|^{2(1+p-q)}. \qquad \square$$

3.2 NCG phase: restarted iterations ($k \in \mathcal{R}$)

Lemma 2. *Let A1 hold and $k \in \mathcal{R}$ with $\|g_k\| > 0$. Then:*

$$f(x_k) - f(x_{k+1}) \geq c_{\mathcal{R}}\, \|g_k\|^2. \tag{13}$$

Proof. For restarted iterations, $d_k = -g_k$, corresponding to $\sigma = \kappa = 1$ and $p = q = 1$ in Lemma 1. The stepsize $\alpha = 1/L_1$ gives:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L_1}\|g_k\|^2. \qquad \square$$

3.3 Negative curvature phase

Lemma 3. *Let A1–A2 hold and $-\lambda_k \geq \varepsilon_H > 0$. Then the step $x_{k+1} = x_k + \frac{2\lambda_k}{L_2} e_k$ satisfies:*

$$f(x_k) - f(x_{k+1}) \geq \frac{2|\lambda_k|^3}{3L_2^2} \geq \frac{2\varepsilon_H^3}{3L_2^2}. \tag{14}$$

Proof. Since $\nabla^2 f$ is L_2-Lipschitz, a third-order Taylor expansion gives:

$$f(x_{k+1}) \leq f(x_k) + \langle g_k, x_{k+1} - x_k\rangle + \frac{1}{2}\langle \nabla^2 f(x_k)(x_{k+1} - x_k), x_{k+1} - x_k\rangle + \frac{L_2}{6}\|x_{k+1} - x_k\|^3.$$

Since e_k is a unit eigenvector with eigenvalue λ_k and $\langle g_k, e_k\rangle \leq 0$, $\lambda_k < 0$, the first-order term $\frac{2\lambda_k}{L_2}\langle g_k, e_k\rangle \geq 0$. Dropping it:

$$f(x_{k+1}) \leq f(x_k) + \frac{1}{2} \cdot \frac{2\lambda_k^2}{L_2} \cdot \frac{2\lambda_k}{L_2} + \frac{L_2}{6}\left(\frac{2|\lambda_k|}{L_2}\right)^3 = f(x_k) - \frac{2|\lambda_k|^3}{3L_2^2}. \qquad \square$$

Remark 1. *All three lemmas guarantee $f(x_{k+1}) < f(x_k)$. This monotone decrease property ensures iterates remain within sublevel sets of f.*

4 Complexity on Arbitrary Nonconvex Functions

We first analyze Algorithm 1 on a general nonconvex function satisfying A1–A3, without assuming the strict saddle property.

Theorem 1. *Under A1–A3, Algorithm 1 reaches a point satisfying $\|\nabla f(x_\star)\| \leq \varepsilon_g$ and $-\lambda_{\min}(\nabla^2 f(x_\star)) \leq \varepsilon_H$ in at most K iterations, where:*

$$K \leq \frac{\Delta_0}{c_{\mathcal{R}}}\, \varepsilon_g^{-2} + \frac{\Delta_0}{c_{\mathcal{N}}}\, \varepsilon_g^{-2(1+p-q)} + \frac{3L_2^2\, \Delta_0}{2}\, \varepsilon_H^{-3}. \tag{15}$$

Proof. Let K be the first index such that the stopping condition holds. Every iteration $k < K$ is either an NCG step ($\|g_k\| \geq \varepsilon_g$) or an NCD step ($\|g_k\| < \varepsilon_g$, $-\lambda_k \geq \varepsilon_H$). Among NCG iterations, each is either restarted or non-restarted.

By Lemma 2, each restarted NCG iteration decreases f by at least $c_{\mathcal{R}}\, \varepsilon_g^2$. By Lemma 1, each non-restarted NCG iteration decreases f by at least $c_{\mathcal{N}}\, \varepsilon_g^{2(1+p-q)}$. By Lemma 3, each NCD iteration decreases f by at least $\frac{2\varepsilon_H^3}{3L_2^2}$.

Telescoping and using A3:

$$\Delta_0 \geq K_{\mathcal{R}}^1 \cdot c_{\mathcal{R}}\, \varepsilon_g^2 + K_{\mathcal{N}}^1 \cdot c_{\mathcal{N}}\, \varepsilon_g^{2(1+p-q)} + K^2 \cdot \frac{2\varepsilon_H^3}{3L_2^2}.$$

Since all terms are positive, each is individually bounded by Δ_0. Summing the resulting bounds on $K_{\mathcal{R}}^1$, $K_{\mathcal{N}}^1$, and K^2 gives (15). $\square$

Remark 2. *The bound* (15) *is conservative because each group uses the full budget Δ_0 independently. The NCG terms match those of Theorem 3.1 in [1] (up to constants due to the absence of line search), while the NCD term follows from [3].*

5 Complexity on Strict Saddle Functions

We now specialize to functions satisfying the strict saddle property, which provides additional structure near stationary points and allows a finer complexity analysis than the generic nonconvex setting of Section 4.

5.1 The strict saddle property

Definition 1 ([2]). *A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is $(\zeta, \beta, \gamma, \delta)$-*strict saddle *if, for every $x \in \mathbb{R}^n$, at least one of the following holds:*

1. ***(Large gradient)*** $\|\nabla f(x)\| \geq \zeta$*,*
2. ***(Negative curvature)*** $\lambda_{\min}(\nabla^2 f(x)) \leq -\beta$*,*
3. ***(Near a local minimizer)*** *there exists a local minimizer $x^\star$ with $\|x - x^\star\| \leq \delta$, and $f \restriction_{B(x^\star, 2\delta)}$ is γ-strongly convex.*

Setting $\varepsilon_g \leq \zeta$ and $\varepsilon_H \leq \beta$ in Algorithm 1 ensures that, upon termination, condition (3) holds: the output lies near a local minimizer. The three conditions define three mutually exclusive *regimes* at each iterate, which we analyze separately.

Notation. Let $K_{\mathcal{R}}^j$ and $K_{\mathcal{N}}^j$ denote the number of restarted and non-restarted NCG iterations in Regime j respectively, K^2 the number of negative curvature iterations, and $K = K_{\mathcal{R}}^1 + K_{\mathcal{N}}^1 + K^2 + K_{\mathcal{R}}^3 + K_{\mathcal{N}}^3$ the total iteration count. We write $\Delta_k := f(x_k) - f(x^\star)$ for the suboptimality at iterate k, where $x^\star$ denotes the local minimizer associated with the current regime. Since all per-iteration decreases are non negative and telescope to at most Δ_0, each regime's iteration count is bounded by dividing Δ_0 by the minimum per-iteration decrease in that regime.

5.2 Regime 1: Large gradient ($\|g_k\| \geq \zeta$)

In Regime 1, every iterate satisfies $\|g_k\| \geq \zeta$. Lemmas 2 and 1 give minimum per-iteration decreases of $c_{\mathcal{R}}\zeta^2$ and $c_{\mathcal{N}}\zeta^{2(1+p-q)}$ for restarted and non-restarted iterations respectively. Telescoping against Δ_0:

$$K^1 \;=\; K_{\mathcal{R}}^1 + K_{\mathcal{N}}^1 \;\leq\; \frac{\Delta_0}{c_{\mathcal{R}}}\,\zeta^{-2} \;+\; \frac{\Delta_0}{c_{\mathcal{N}}}\,\zeta^{-2(1+p-q)}. \tag{16}$$

5.3 Regime 2: Negative curvature ($\lambda_{\min}(\nabla^2 f(x_k)) \leq -\beta$)

In Regime 2, every iterate satisfies $\lambda_{\min}(\nabla^2 f(x_k)) \leq -\beta$. Lemma 3 guarantees a minimum per-iteration decrease of $2\beta^3/(3L_2^2)$. Telescoping against Δ_0:

$$K^2 \;\leq\; \frac{3L_2^2\,\Delta_0}{2\beta^3}. \tag{17}$$

5.4 Regime 3: Near a local minimizer

Throughout this section we assume $x_k \in B(x^\star, \delta)$ and that f is γ-strongly convex in $B(x^\star, 2\delta)$. Both the smoothness upper bound (7) and the PL inequality (8) are available. We write $\Delta_k := f(x_k) - f(x^\star)$ for the suboptimality at iterate k.

The gradient stopping criterion $\|g_k\| \leq \varepsilon_g$ is equivalent to requiring $\Delta_k \leq \varepsilon$ via the smoothness bound (7), where we set:

$$\varepsilon \;:=\; \frac{\varepsilon_g^2}{2L_1}. \tag{18}$$

This conversion is used throughout the analysis of Regime 3.

5.4.1 Entry conditions and the role of δ

At the first Regime-3 iterate k_0, the strict saddle property gives $\|x_{k_0} - x^\star\| \le \delta$. Applying the descent lemma from $x^\star$ with $\nabla f(x^\star) = 0$:

$$\Delta_{k_0} \;\le\; \tfrac{L_1}{2}\delta^2. \tag{19}$$

Strong convexity at x_{k_0} gives the matching lower bound:

$$\Delta_{k_0} \;\ge\; \tfrac{\gamma}{2}\|x_{k_0} - x^\star\|^2. \tag{20}$$

Combining with (7) and (8):

$$\gamma\|x_{k_0} - x^\star\| \;\le\; \|g_{k_0}\| \;\le\; L_1\delta. \tag{21}$$

5.4.2 Restarted iterations in Regime 3 ($k \in \mathcal{R}^3$)

For restarted iterations $d_k = -g_k$ with stepsize $\alpha = 1/L_1$, the descent lemma and PL inequality give the standard gradient-descent contraction:

$$\Delta_{k+1} \;\le\; \Big(1 - \frac{\gamma}{L_1}\Big)\Delta_k. \tag{22}$$

Telescoping and requiring $\Delta_K \le \varepsilon_g^2/(2L_1)$:

$$K^3_{\mathcal{R}} \;\le\; \frac{L_1}{\gamma}\log\frac{L_1^2\delta^2}{\varepsilon_g^2}. \tag{23}$$

5.4.3 Non-restarted iterations in Regime 3 ($k \in \mathcal{N}^3$)

We present two complexity results for non-restarted iterations. Theorem 2 uses a Lyapunov function argument and gives the sharpest possible convergence classification, including finite termination. Theorem **??** uses a uniform geometric contraction and produces a bound directly comparable to [1].
Lemma 1 gives, for every $k \in \mathcal{N}^3$:

$$\Delta_k - \Delta_{k+1} \;\ge\; c_{\mathcal{N}}\|g_k\|^{2r}, \qquad r := 1 + p - q, \tag{24}$$

where $c_{\mathcal{N}} = \sigma^2/(2L_1\kappa^2)$. Applying the PL inequality $\|g_k\|^2 \ge 2\gamma\Delta_k$ to (24) yields the fundamental recurrence:

$$\Delta_{k+1} \;\le\; \Delta_k - C\Delta_k^r, \qquad C \;:=\; c_{\mathcal{N}}(2\gamma)^r = \frac{\sigma^2(2\gamma)^r}{2L_1\kappa^2}. \tag{25}$$

Both theorems below depart from (25).

Theorem 2 (Lyapunov function bound)**.** *Let f satisfy A1–A2 and the $(\zeta, \beta, \gamma, \delta)$-strict saddle property. Let $\{x_k\}_{k\in\mathcal{N}^3}$ be generated by Algorithm 1 with stepsize $\alpha^\star = \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q}$, and let $r := 1 + p - q$ and $C := \frac{\sigma^2(2\gamma)^r}{2L_1\kappa^2}$.*

(i) ***Linear convergence*** *($p = q$, $r = 1$):*

$$K^3_{\mathcal{N},\,p=q} \;\le\; \frac{L_1\kappa^2}{\sigma^2\gamma}\log\frac{L_1^2\delta^2}{\varepsilon_g^2}. \tag{26}$$

(ii) ***Sublinear convergence*** *($p > q$, $r > 1$):*

$$K^3_{\mathcal{N},\,p>q} \;\le\; \frac{2L_1\kappa^2}{\sigma^2(2\gamma)^r(p-q)}\left[\left(\frac{\varepsilon_g^2}{2L_1}\right)^{q-p} - \left(\frac{L_1\delta^2}{2}\right)^{q-p}\right]. \tag{27}$$

(iii) ***Finite termination*** *($p < q$, $r < 1$):*

$$K^3_{\mathcal{N},\, p<q} \;\le\; \left(\frac{L_1\delta^2}{2}\right)^{q-p} \frac{2L_1\kappa^2}{\sigma^2(2\gamma)^r(q-p)}. \tag{28}$$

This bound is independent of ε_g: the algorithm reaches $f(x_{K^\star}) = f(x^\star)$ exactly in finitely many non-restarted steps.

Proof. All three cases depart from recurrence (25).

Case (i): $r = 1$. Recurrence (25) becomes $\Delta_{k+1} \le (1 - C)\Delta_k$, a geometric contraction with $C = \sigma^2\gamma/(L_1\kappa^2) \in (0, 1)$. Telescoping over $K^3_{\mathcal{N}}$ iterations and using $-\log(1 - C) \ge C$:

$$K^3_{\mathcal{N}} \;\le\; \frac{1}{C}\log\frac{\Delta_{k_0}}{\varepsilon} \;\le\; \frac{L_1\kappa^2}{\sigma^2\gamma}\log\frac{L_1^2\delta^2}{\varepsilon_g^2},$$

where we substituted (19) and $\varepsilon = \varepsilon_g^2/(2L_1)$.

Case (ii): $r > 1$. Define the Lyapunov function $\phi_k := \Delta_k^{1-r}$, where $1 - r < 0$. Since $t \mapsto t^{1-r}$ is *convex* for $r > 1$, raising both sides of (25) to the power $1 - r < 0$ flips the inequality:

$$\phi_{k+1} \;\ge\; \Delta_k^{1-r}(1 - C\Delta_k^{r-1})^{1-r} \;=\; \phi_k(1 - C\phi_k^{-1})^{1-r}.$$

Applying the convexity bound $(1 - x)^{1-r} \ge 1 + (r - 1)x$ for $x \in (0, 1)$ (which holds since $t \mapsto t^{1-r}$ is convex):

$$\phi_{k+1} \;\ge\; \phi_k\bigl(1 + (r - 1)C\phi_k^{-1}\bigr) = \phi_k + (r - 1)C.$$

We first verify that $a := C\Delta_k^{r-1} \in (0, 1)$. Positivity is immediate since $C > 0$ and $\Delta_k > 0$. For the upper bound, suppose $a \ge 1$: then the right-hand side of (25) satisfies $\Delta_k - C\Delta_k^r = \Delta_k(1 - a) \le 0$, giving $\Delta_{k+1} \le f(x^\star)$. Since $x_{k+1} \in B(x^\star, 2\delta)$ where f is strongly convex, $f(x_{k+1}) \ge f(x^\star)$, so $\Delta_{k+1} = 0$: the algorithm has already converged. Hence for any non-terminated run, $a \in (0, 1)$. Telescoping over $K^3_{\mathcal{N}}$ iterations:

$$\phi_K \;\ge\; \phi_0 + K^3_{\mathcal{N}}(r - 1)C.$$

Requiring $\Delta_K \le \varepsilon$ with $1 - r < 0$ gives $\phi_K \ge \varepsilon^{1-r}$, hence:

$$K^3_{\mathcal{N}} \;\le\; \frac{\varepsilon^{1-r} - \phi_0}{(r - 1)C}.$$

Substituting $\phi_0 = \Delta_{k_0}^{1-r} \ge (L_1\delta^2/2)^{1-r}$, $C = \sigma^2(2\gamma)^r/(2L_1\kappa^2)$, and $\varepsilon = \varepsilon_g^2/(2L_1)$ yields (27).

Case (iii): $r < 1$, $r > 0$. Define the Lyapunov function $\phi_k := \Delta_k^{1-r}$, where $1 - r > 0$. The function $t \mapsto t^{1-r}$ is *concave* for $0 < r < 1$. Raising both sides of (25) to the power $1 - r > 0$ preserves the inequality:

$$\phi_{k+1} \;\le\; \phi_k(1 - C\phi_k^{-1})^{1-r}.$$

Applying the concavity bound $(1 - x)^{1-r} \le 1 + (r - 1)x$ for $x \in (0, 1)$:

$$\phi_{k+1} \;\le\; \phi_k\bigl(1 + (r - 1)C\phi_k^{-1}\bigr) = \phi_k - (1 - r)C.$$

Hence ϕ_k decreases by the fixed amount $(1 - r)C > 0$ at every iteration. Telescoping:

$$\phi_K \;\le\; \phi_0 - K^3_{\mathcal{N}}(1 - r)C.$$

Since $\phi_K = \Delta_K^{1-r} \ge 0$:

$$K^3_{\mathcal{N}} \;\le\; \frac{\phi_0}{(1 - r)C}.$$

At $K^\star = \lceil \phi_0/((1-r)C) \rceil$, the recursion requires $\phi_{K^\star} \le 0$. Since $\phi_{K^\star} \ge 0$, the only consistent value is $\phi_{K^\star} = 0$, i.e. $f(x_{K^\star}) = f(x^\star)$ exactly. Substituting $\phi_0 \le (L_1\delta^2/2)^{1-r}$ and $C = \sigma^2(2\gamma)^r/(2L_1\kappa^2)$ yields (28). $\square$

Remark 3 (Extension to $r \le 0$)**.** *When $q > 1+p$, i.e. $r \le 0$, the concavity argument above no longer applies since $t \mapsto t^{1-r}$ becomes convex. However, since $r \le 0$ and Δ_k is decreasing, Δ_k^r is increasing, so $\Delta_k^r \ge \Delta_{k_0}^r$ for all $k \ge k_0$. Recurrence* (25) *then gives $\Delta_{k+1} \le \Delta_k - C\Delta_{k_0}^r$: a constant decrease of $C\Delta_{k_0}^r$ per step. The algorithm therefore reaches $f(x^\star)$ exactly in at most $\lceil \Delta_{k_0}/(C\Delta_{k_0}^r)\rceil = \lceil \Delta_{k_0}^{1-r}/C\rceil$ iterations. It yields finite termination with a different constant than* (28)*, but the same qualitative conclusion.*

Remark 4 (Comparison with gradient descent)**.** *Bound* (26) *matches the restarted rate* (23) *up to the factor $\sigma^2/\kappa^2 \le 1$, recovering the gradient descent baseline for $p = q$. Bound* (28) *establishes that, for any $p < q$, the non-restarted NCG phase reaches the* exact *minimizer in finitely many steps which an improvement over gradient descent, which can only guarantee $\Delta_k \le \varepsilon$ for any fixed $\varepsilon > 0$.*

5.4.4 Restarted iterations in Regime 3 ($k \in \mathcal{R}^3$)

From Lemma 2 and the PL inequality:

$$f(x_{k+1}) - f(x^\star) \le \left(1 - \frac{\gamma}{L_1}\right)(f(x_k) - f(x^\star)).$$

Iterating and requiring $f(x_K) - f(x^\star) \le \varepsilon$, using $-\log(1-x) \ge x$ for $x \in (0,1)$:

$$K_{\mathcal{R}}^3 \le \frac{L_1}{\gamma}\log\frac{L_1\delta^2}{2\varepsilon}. \tag{29}$$

5.5 Overall complexity

Theorem 3. *Under A1–A3, suppose f is $(\zeta,\beta,\gamma,\delta)$-strict saddle. Then Algorithm 1 with $\varepsilon_g = \zeta$ and $\varepsilon_H = \beta$ reaches a point near a local minimizer in at most K iterations, where:*

$$K \;\le\; \underbrace{\frac{\Delta_0}{c_{\mathcal{R}}}\zeta^{-2} + \frac{\Delta_0}{c_{\mathcal{N}}}\zeta^{-2(1+p-q)}}_{\textit{Regime 1}} + \underbrace{\frac{3L_2^2\,\Delta_0}{2\beta^3}}_{\textit{Regime 2}} + \underbrace{\frac{L_1}{\gamma}\log\frac{L_1^2\delta^2}{\varepsilon_g^2} + K_{\mathcal{N}}^3}_{\textit{Regime 3}}, \tag{30}$$

with $r := 1+p-q$ and $K_{\mathcal{N}}^3$ given by Theorem 2:

$$K_{\mathcal{N}}^3 \;=\; \begin{cases} \dfrac{L_1\kappa^2}{\sigma^2\gamma}\log\dfrac{L_1^2\delta^2}{\varepsilon_g^2} & \textit{if } p = q,\\[2ex] \dfrac{2L_1\kappa^2}{\sigma^2(2\gamma)^r(p-q)}\left[\left(\dfrac{\varepsilon_g^2}{2L_1}\right)^{q-p} - \left(\dfrac{L_1\delta^2}{2}\right)^{q-p}\right] & \textit{if } p > q,\\[2ex] \left(\dfrac{L_1\delta^2}{2}\right)^{q-p}\dfrac{2L_1\kappa^2}{\sigma^2(2\gamma)^r(q-p)} & \textit{if } p < q. \end{cases}$$

Proof. Sum the bounds from (16), (17), (23), and Theorem 2. Each regime bound uses Δ_0 independently, which is conservative: the true budget satisfies $\Delta_0^1 + \Delta_0^2 + \Delta_0^3 \le \Delta_0$, so the individual bounds could in principle be tightened if the per-regime budgets were known a priori. □

Remark 5. *The restarted terms ζ^{-2} in Regime 1 and $\frac{L_1}{\gamma}\log\frac{1}{\varepsilon_g}$ in Regime 3 match the gradient descent baseline. The non-restarted Regime 3 term $K_{\mathcal{N}}^3$ is the novel contribution: for $p < q$ it is independent of ε_g, so the overall bound improves over gradient descent whenever the algorithm spends a significant fraction of its iterations in the non-restarted Regime 3 phase.*

6 Regime 3 Analysis

6.1 Staying in Strong-Convexity Region (proof 1 using strongly convex + descent lemma)

Proposition 1. *Suppose f is γ-strongly convex in $B(x^\star, 2\delta) := \{x : \|x - x^\star\| \le 2\delta\}$ and A1 holds globally. Let k_0 be the first iteration at which $x_{k_0} \in B(x^\star,\delta)$. If $L_1 \le 4\gamma$, then $x_k \in B(x^\star, 2\delta)$ for all $k \ge k_0$.*

Proof. We proceed by induction on $k \geq k_0$.

Base case ($k = k_0$). By assumption, $x_{k_0} \in B(x^\star, \delta)$. Since $\delta < 2\delta$, we have $x_{k_0} \in B(x^\star, 2\delta)$.

Inductive step. Assume $x_k \in B(x^\star, 2\delta)$ for some $k \geq k_0$. We want to show $x_{k+1} \in B(x^\star, 2\delta)$.

Step 1: $f(x_{k+1}) \leq f(x_{k_0})$. Since A1 holds globally, the descent lemma (6) applies at every iterate, regardless of its location. As established in Lemmas 1, 2, and 3, every iteration of Algorithm 1 (whether non-restarted, restarted, or negative curvature) produces strict decrease:

$$f(x_{k+1}) < f(x_k) \qquad \text{for all } k \geq 0.$$

Applying this repeatedly from k_0 to k:

$$f(x_{k+1}) < f(x_k) < f(x_{k-1}) < \cdots < f(x_{k_0}).$$

In particular:

$$f(x_{k+1}) \leq f(x_{k_0}). \tag{31}$$

Step 2: Upper bound on $f(x_{k_0})$. From the entry condition (Section 5.4.1), since $x_{k_0} \in B(x^\star, \delta)$ and $\nabla f(x^\star) = 0$, the descent lemma with $x = x^\star$ and $y = x_{k_0}$ gives:

$$f(x_{k_0}) \leq f(x^\star) + \frac{L_1}{2}\|x_{k_0} - x^\star\|^2 \leq f(x^\star) + \frac{L_1}{2}\delta^2.$$

Combining with (31):

$$f(x_{k+1}) \leq f(x^\star) + \frac{L_1}{2}\delta^2. \tag{32}$$

Step 3: Lower bound on f at the boundary of $B(x^\star, 2\delta)$. Suppose for contradiction that $\|x_{k+1} - x^\star\| > 2\delta$, i.e., $x_{k+1} \notin B(x^\star, 2\delta)$. Consider any point $\bar{x}$ on the boundary of $B(x^\star, 2\delta)$, i.e., $\|\bar{x} - x^\star\| = 2\delta$. Since f is γ-strongly convex in $B(x^\star, 2\delta)$ and $\nabla f(x^\star) = 0$:

$$f(\bar{x}) \geq f(x^\star) + \nabla f(x^\star)^T(\bar{x} - x^\star) + \frac{\gamma}{2}\|\bar{x} - x^\star\|^2 = f(x^\star) + \frac{\gamma}{2}(2\delta)^2 = f(x^\star) + 2\gamma\delta^2.$$

This means: to leave $B(x^\star, 2\delta)$, the function value must exceed $f(x^\star) + 2\gamma\delta^2$, since f increases (by strong convexity) as we move away from $x^\star$. By continuity of f, if x_{k+1} is outside $B(x^\star, 2\delta)$, there exists a point on the path from x_k to x_{k+1} where f equals $f(x^\star) + 2\gamma\delta^2$. Since $f(x_{k+1}) < f(x_k)$ and $x_k \in B(x^\star, 2\delta)$ (inductive hypothesis), the iterate must cross the boundary, so:

$$f(x_{k+1}) \geq f(x^\star) + 2\gamma\delta^2. \tag{33}$$

Step 4: Contradiction. Combining (32) and (33):

$$f(x^\star) + 2\gamma\delta^2 \leq f(x_{k+1}) \leq f(x^\star) + \frac{L_1}{2}\delta^2.$$

Subtracting $f(x^\star)$ from all terms:

$$2\gamma\delta^2 \leq \frac{L_1}{2}\delta^2.$$

Dividing by $\delta^2 > 0$:

$$2\gamma \leq \frac{L_1}{2} \iff 4\gamma \leq L_1.$$

This contradicts the assumption $L_1 \leq 4\gamma$. Therefore $x_{k+1} \in B(x^\star, 2\delta)$, completing the inductive step. $\square$

6.2 Staying in Strong-Convexity Region (proof 2 using step length bound)

We provide an alternative proof that iterates remain in $B(x^\star, 2\delta)$, using control of the step length. This approach replaces the condition $L_1 \le 4\gamma$ with a condition on δ.

Proposition 2. *Suppose f is γ-strongly convex in $B(x^\star, 2\delta)$ and A1 holds globally. Let k_0 be the first iteration at which $x_{k_0} \in B(x^\star, \delta)$. Suppose one of the following holds:*

1. $p = q$,
2. $p > q$ *and* $\delta \le \left(\frac{\kappa}{\sigma}\right)^{1/(p-q)} \frac{1}{L_1}$,
3. $p < q$ *and* $\delta \ge \left(\frac{\sigma}{\kappa}\right)^{1/(q-p)} \frac{1}{L_1}$.

Then $x_k \in B(x^\star, 2\delta)$ for all $k \ge k_0$.

Proof. We proceed by induction on $k \ge k_0$. The base case $x_{k_0} \in B(x^\star, \delta) \subset B(x^\star, 2\delta)$ holds by assumption. Assume $x_k \in B(x^\star, 2\delta)$ for some $k \ge k_0$. We show $x_{k+1} \in B(x^\star, 2\delta)$ by bounding the step length for each iteration type.

6.2.1 Non-restarted case ($k \in \mathcal{N}$)

Recall that $x_{k+1} = x_k + \alpha_k d_k$, where $d_k = -g_k + \beta_k d_{k-1}$ (with $d_0 = -g_0$). Under non-restarted iterations, condition (9) is not triggered, so:

$$g_{k+1}^T d_{k+1} \le -\sigma \|g_{k+1}\|^{1+p} \quad \text{and} \quad \|d_{k+1}\| \le \kappa \|g_{k+1}\|^q.$$

From the entry condition (Section 5.4.1), for all $k \ge k_0$ such that $x_k \in B(x^\star, \delta)$:

$$\|g_k\| \le L_1 \delta.$$

The step length satisfies:

$$\|x_{k+1} - x_k\| = \alpha_k \|d_k\| \le \alpha_k \kappa \|g_k\|^q.$$

Substituting the optimal stepsize $\alpha_k^\star = \frac{\sigma}{L_1 \kappa^2} \|g_k\|^{1+p-2q}$:

$$\begin{aligned}\|x_{k+1} - x_k\| &\le \frac{\sigma}{L_1 \kappa^2} \|g_k\|^{1+p-2q} \cdot \kappa \|g_k\|^q \\ &= \frac{\sigma}{L_1 \kappa} \|g_k\|^{1+p-q}.\end{aligned}$$

Using $\|g_k\| \le L_1 \delta$:

$$\|x_{k+1} - x_k\| \le \frac{\sigma}{L_1 \kappa} (L_1 \delta)^{1+p-q} = \frac{\sigma}{\kappa} L_1^{p-q} \, \delta^{1+p-q}. \tag{34}$$

We want $\|x_{k+1} - x_k\| \le \delta$, i.e.:

$$\frac{\sigma}{\kappa} L_1^{p-q} \, \delta^{1+p-q} \le \delta \iff \frac{\sigma}{\kappa} (L_1 \delta)^{p-q} \le 1.$$

Three cases arise depending on the sign of $p - q$:

Case $p = q$: The condition becomes $\frac{\sigma}{\kappa} \le 1$, which always holds since $\sigma \le 1 \le \kappa$.

Case $p > q$ ($p - q > 0$): The condition becomes:

$$(L_1 \delta)^{p-q} \le \frac{\kappa}{\sigma} \iff \delta \le \left(\frac{\kappa}{\sigma}\right)^{1/(p-q)} \frac{1}{L_1}.$$

This is a condition for δ small enough, which is satisfied for sufficiently small δ.

Case $p < q$ ($p - q < 0$): Since $p - q < 0$, we have $(L_1\delta)^{p-q} = \frac{1}{(L_1\delta)^{q-p}} \to \infty$ as $\delta \to 0$. The condition becomes:

$$(L_1\delta)^{p-q} \le \frac{\kappa}{\sigma} \iff \delta^{p-q} \le \frac{\kappa}{\sigma} L_1^{q-p} \iff \delta \ge \left(\frac{\sigma}{\kappa}\right)^{1/(q-p)} \frac{1}{L_1}.$$

This is a condition for δ *large enough*: the neighborhood cannot be too small. This creates a tension with the finite termination result (Case 3 in Section 5.4), which also requires $p < q$.

6.2.2 Restarted case ($k \in \mathcal{R}$)

For restarted iterations, $d_k = -g_k$ and $\alpha_k = 1/L_1$. The step length is:

$$\|x_{k+1} - x_k\| = \frac{1}{L_1}\|g_k\| \le \frac{1}{L_1} \cdot L_1\delta = \delta.$$

No condition on any parameter is needed: the step length is always bounded by δ.

By the triangle inequality, for both cases:

$$\|x_{k+1} - x^\star\| \le \|x_{k+1} - x_k\| + \|x_k - x^\star\| \le \delta + \delta = 2\delta.$$

Therefore $x_{k+1} \in B(x^\star, 2\delta)$, completing the induction. □

6.3 Are we staying in Regime 3 (Same proof technique as in 6.1)

By using the exact same proof technique as in 6.1, but replacing $B(x^\star, 2\delta)$ by $B(x^\star, \delta)$, we find the following condition in the contradiction:

$$L_1 \ge \gamma. \tag{35}$$

This is always true, meaning that we can prove that we stay in Regime 3 only for $L_1 = \gamma$, which is the most well-posed a problem can be. This proof technique only allows to prove that we stay in Regime 3 once we enter it for perfect quadratic (i.e. condition number is 1) functions, that is function of the form $f(x) = x^T Ax + b^T x + c$ with $A = \gamma I$. This is a well-known not interesting result.

6.4 Staying in Regime 3 (Using gradient norm bounds)

We attempt to prove that iterates eventually enter and stay in $B(x^\star, \delta)$, not just $B(x^\star, 2\delta)$, by showing that gradient norms shrink, which in turn shrinks step lengths and distances to $x^\star$.

6.4.1 Restarted case ($k \in \mathcal{R}$)

For restarted iterations, $d_k = -g_k$, $\alpha_k = 1/L_1$, so $x_{k+1} - x_k = -g_k/L_1$. We expand the gradient at x_{k+1} using a Taylor expansion:

$$\nabla f(x_{k+1}) = \nabla f(x_k) + \nabla^2 f(x_k)(x_{k+1} - x_k) + R_k,$$

where R_k is the remainder. Substituting $x_{k+1} - x_k = -g_k/L_1$:

$$g_{k+1} = g_k - \frac{1}{L_1}\nabla^2 f(x_k) g_k + R_k = \left(I - \frac{\nabla^2 f(x_k)}{L_1}\right) g_k + R_k.$$

Bounding the remainder. We write:

$$R_k = g_{k+1} - g_k - \nabla^2 f(x_k)(x_{k+1} - x_k) = \int_0^1 \left[\nabla^2 f(x_k + t(x_{k+1} - x_k)) - \nabla^2 f(x_k)\right](x_{k+1} - x_k)\, dt.$$

Taking norms and using A2 ($\nabla^2 f$ is L_2-Lipschitz):

$$\|R_k\| \le \int_0^1 \|\nabla^2 f(x_k + t(x_{k+1} - x_k)) - \nabla^2 f(x_k)\| \cdot \|x_{k+1} - x_k\|\, dt \le \int_0^1 L_2\, t\, \|x_{k+1} - x_k\|^2\, dt = \frac{L_2}{2}\|x_{k+1} - x_k\|^2.$$

Since $\|x_{k+1} - x_k\| = \|g_k\|/L_1$:

$$\|R_k\| \le \frac{L_2}{2L_1^2}\|g_k\|^2.$$

Gradient contraction. Taking norms in the Taylor expansion:

$$\|g_{k+1}\| \leq \left\| I - \frac{\nabla^2 f(x_k)}{L_1} \right\| \|g_k\| + \|R_k\|.$$

In the strongly convex region $B(x^\star, 2\delta)$, the eigenvalues of $\nabla^2 f(x_k)$ lie in $[\gamma, L_1]$, so the eigenvalues of $I - \nabla^2 f(x_k)/L_1$ lie in $[0, 1 - \gamma/L_1]$. Therefore:

$$\|g_{k+1}\| \leq \left(1 - \frac{\gamma}{L_1}\right) \|g_k\| + \frac{L_2}{2L_1^2} \|g_k\|^2. \tag{36}$$

Condition for gradient shrinkage. We need the right-hand side of (36) to be strictly less than $\|g_k\|$:

$$\left(1 - \frac{\gamma}{L_1}\right) \|g_k\| + \frac{L_2}{2L_1^2} \|g_k\|^2 < \|g_k\| \iff \frac{L_2}{2L_1^2} \|g_k\| < \frac{\gamma}{L_1}.$$

In the strongly convex region, $\|g_k\| \leq L_1 \delta$ (from Section 5.4.1). Substituting:

$$\frac{L_2}{2L_1^2} \cdot L_1 \delta < \frac{\gamma}{L_1} \iff \frac{L_2}{2L_1} \delta < \frac{\gamma}{L_1} \iff \delta < \frac{2\gamma}{L_2}.$$

Under this condition, gradients shrink at every iteration.

Step length and distance to $x^\star$. Since gradients shrink, step lengths also shrink:

$$\|x_{k+1} - x_k\| = \frac{\|g_k\|}{L_1} \leq \delta.$$

From strong monotonicity ($\|g_k\| \geq \gamma \|x_k - x^\star\|$, see Section 5.4.1):

$$\|x_k - x^\star\| \leq \frac{\|g_k\|}{\gamma}.$$

By the triangle inequality:

$$\|x_{k+1} - x^\star\| \leq \|x_{k+1} - x_k\| + \|x_k - x^\star\| \leq \frac{\|g_k\|}{L_1} + \frac{\|g_k\|}{\gamma} = \|g_k\| \left(\frac{1}{L_1} + \frac{1}{\gamma}\right).$$

For this to be $\leq \delta$, we would need:

$$L_1 \delta \left(\frac{1}{L_1} + \frac{1}{\gamma}\right) \leq \delta \iff 1 + \frac{L_1}{\gamma} \leq 1 \iff \frac{L_1}{\gamma} \leq 0,$$

which is impossible. Therefore, the first step after entering $B(x^\star, \delta)$ may leave this ball.

Eventual convergence to $B(x^\star, \delta)$. Although the first step may leave $B(x^\star, \delta)$, the iterate remains in $B(x^\star, 2\delta)$ (from Proposition 1 or 2), so the gradient contraction (36) continues to apply. Under $\delta < 2\gamma/L_2$, the effective contraction rate is:

$$\bar{\rho} := 1 - \frac{\gamma}{L_1} + \frac{L_2 \delta}{2L_1} < 1.$$

Iterating from k_0:

$$\|g_{k_0+m}\| \leq \bar{\rho}^m \|g_{k_0}\| \leq \bar{\rho}^m L_1 \delta.$$

The distance to $x^\star$ satisfies:

$$\|x_{k_0+m} - x^\star\| \leq \frac{\|g_{k_0+m}\|}{\gamma} \leq \frac{\bar{\rho}^m L_1 \delta}{\gamma}.$$

The step length at iteration m:

$$\|x_{k_0+m+1} - x_{k_0+m}\| \leq \frac{\|g_{k_0+m}\|}{L_1} \leq \bar{\rho}^m \delta.$$

By the triangle inequality:

$$\|x_{k_0+m+1} - x^\star\| \leq \bar{\rho}^m \delta + \frac{\bar{\rho}^m L_1 \delta}{\gamma} = \bar{\rho}^m \delta \left(1 + \frac{L_1}{\gamma}\right).$$

This is $\leq \delta$ when:

$$\bar{\rho}^m \left(1 + \frac{L_1}{\gamma}\right) \leq 1.$$

Taking logarithms and using $-\log \bar{\rho} \geq \frac{2\gamma - L_2\delta}{2L_1}$:

$$m^\star_{\mathcal{R}} = \left\lceil \frac{2L_1 \log(1 + L_1/\gamma)}{2\gamma - L_2\delta} \right\rceil. \tag{37}$$

After $m^\star_{\mathcal{R}}$ restarted iterations in the strongly convex region, all subsequent iterates remain in $B(x^\star, \delta)$.

Summary. Under $\delta < 2\gamma/L_2$:

1. Gradient norms shrink geometrically: $\|g_{k+1}\| \leq \bar{\rho}\|g_k\|$ with $\bar{\rho} < 1$.
2. Iterates stay in $B(x^\star, 2\delta)$ from the first iteration k_0.
3. After a transient phase of $m^\star_{\mathcal{R}}$ iterations (37), iterates enter and stay in $B(x^\star, \delta)$.

6.4.2 Non-Restarted Case ($k \in \mathcal{N}$)

We present two attempts to prove gradient shrinkage for non-restarted iterations. Both fail, for instructive reasons.

Attempt 1: Taylor expansion approach. Recall that for non-restarted iterations, $d_k = -g_k + \beta_k d_{k-1}$ and $\alpha^\star_k = \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q}$. The step is $x_{k+1} - x_k = \alpha_k d_k$, so $x_{k+1} - x_k = \alpha_{k+1} d_{k+1}$ at the next iteration. Expanding the gradient at x_{k+1} by Taylor:

$$g_{k+1} = g_k + \nabla^2 f(x_k)(x_{k+1} - x_k) + R_k = g_k + \alpha_k \nabla^2 f(x_k) d_k + R_k.$$

Substituting $\alpha^\star_k$:

$$g_{k+1} = g_k + \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q} \nabla^2 f(x_k) d_k + R_k.$$

Using $\|g_k\| \leq L_1\delta$:

$$g_{k+1} \leq g_k + \frac{\sigma}{\kappa^2}(L_1\delta)^{p-2q} \nabla^2 f(x_k) d_k + R_k.$$

This does not simplify further. In the restarted case, $d_k = -g_k$ allowed the factorization $g_k + \alpha_k \nabla^2 f(x_k) d_k = (I - \frac{\nabla^2 f(x_k)}{L_1})g_k$, yielding eigenvalue-based contraction. Here, $d_k = -g_k + \beta_k d_{k-1}$ introduces the term $\nabla^2 f(x_k) d_{k-1}$, which couples the current Hessian with the previous direction. The eigenvalue argument does not apply because d_{k-1} is not aligned with any eigenvector of $\nabla^2 f(x_k)$ in general.

Attempt 2: Function value approach. We use the smoothness bound $\|g_{k+1}\|^2 \leq 2L_1(f(x_{k+1}) - f(x^\star))$ and the decrease recursion (**??**):

$$\|g_{k+1}\|^2 \leq 2L_1(f(x_{k+1}) - f(x^\star)) \leq 2L_1(f(x_k) - f(x^\star))\left(1 - C(f(x_k) - f(x^\star))^{r-1}\right).$$

From PL, $f(x_k) - f(x^\star) \leq \frac{\|g_k\|^2}{2\gamma}$. Substituting:

$$\|g_{k+1}\|^2 \leq 2L_1 \cdot \frac{\|g_k\|^2}{2\gamma}\left(1 - C\left(\frac{\|g_k\|^2}{2\gamma}\right)^{r-1}\right) = \frac{L_1}{\gamma}\|g_k\|^2\left(1 - C\left(\frac{\|g_k\|^2}{2\gamma}\right)^{r-1}\right).$$

Substituting $C = \frac{\sigma^2(2\gamma)^r}{2L_1\kappa^2}$:

$$\|g_{k+1}\|^2 \leq \frac{L_1}{\gamma}\|g_k\|^2\left(1 - \frac{\sigma^2(2\gamma)^r}{2L_1\kappa^2} \cdot \frac{\|g_k\|^{2(r-1)}}{(2\gamma)^{r-1}}\right) = \frac{L_1}{\gamma}\|g_k\|^2\left(1 - \frac{\sigma^2\gamma}{L_1\kappa^2}\|g_k\|^{2(r-1)}\right).$$

The contraction factor for $\|g_{k+1}\|^2$ is therefore:

$$\rho_k = \frac{L_1}{\gamma} - \frac{\sigma^2}{\kappa^2}\|g_k\|^{2(r-1)}.$$

For $r = 1$ ($p = q$), this gives $\rho_k = \frac{L_1}{\gamma} - \frac{\sigma^2}{\kappa^2}$. Gradient shrinkage requires $\rho_k < 1$, i.e.:

$$\frac{\sigma^2}{\kappa^2} > \frac{L_1 - \gamma}{\gamma} = \kappa_f - 1.$$

Since $\sigma \leq 1$ and $\kappa \geq 1$, we have $\frac{\sigma^2}{\kappa^2} \leq 1$. But $\kappa_f - 1 > 1$ whenever $\kappa_f > 2$, which is the case for any moderately ill-conditioned problem. The condition is therefore not satisfiable in general.

6.4.3 Non-Restarted Case ($k \in \mathcal{N}$) — Defining $d_k = -H_k g_k$

We write the non-restarted direction as $d_k = -H_k g_k$, where H_k is a positive definite matrix defined implicitly by the conjugate gradient update $d_k = -g_k + \beta_k d_{k-1}$. We denote the eigenvalues of H_k by $\lambda^H_{\min}$ and $\lambda^H_{\max}$.

Step 1: Taylor expansion. Expanding the gradient at x_{k+1}:

$$g_{k+1} = g_k + \nabla^2 f(x_k)(x_{k+1} - x_k) + R_k.$$

Since $x_{k+1} - x_k = \alpha_k d_k = -\alpha_k H_k g_k$:

$$g_{k+1} = g_k - \alpha_k \nabla^2 f(x_k) H_k g_k + R_k = \left(I - \alpha_k H_k \nabla^2 f(x_k)\right) g_k + R_k.$$

Step 2: Bound the remainder. Using A2 and the fundamental theorem of calculus:

$$\|R_k\| \leq \frac{L_2}{2}\|x_{k+1} - x_k\|^2 = \frac{L_2}{2}\|\alpha_k H_k g_k\|^2 \leq \frac{L_2\alpha_k^2(\lambda^H_{\max})^2}{2}\|g_k\|^2.$$

Substituting $\alpha_k = \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q}$:

$$\|R_k\| \leq \frac{L_2(\sigma\lambda^H_{\max})^2}{2L_1^2\kappa^4}\|g_k\|^{2(2+p-2q)}. \tag{38}$$

Step 3: Gradient contraction. Taking norms in the Taylor expansion:

$$\|g_{k+1}\| \leq \left\|I - \alpha_k H_k \nabla^2 f(x_k)\right\| \|g_k\| + \|R_k\|.$$

In $B(x^\star, 2\delta)$, eigenvalues of $\nabla^2 f(x_k)$ lie in $[\gamma, L_1]$ and those of H_k in $[\lambda^H_{\min}, \lambda^H_{\max}]$. The eigenvalues of $I - \alpha_k H_k \nabla^2 f(x_k)$ lie in:

$$\left[1 - \alpha_k\lambda^H_{\max}L_1,\ 1 - \alpha_k\lambda^H_{\min}\gamma\right].$$

Assuming all eigenvalues are nonnegative ($\alpha_k\lambda^H_{\max}L_1 \leq 1$), the operator norm is:

$$\left\|I - \alpha_k H_k \nabla^2 f(x_k)\right\| = 1 - \alpha_k\lambda^H_{\min}\gamma = 1 - \frac{\sigma\lambda^H_{\min}\gamma}{L_1\kappa^2}\|g_k\|^{1+p-2q}.$$

Combining with (38):

$$\|g_{k+1}\| \leq \left(1 - \frac{\sigma\lambda^H_{\min}\gamma}{L_1\kappa^2}\|g_k\|^{1+p-2q}\right)\|g_k\| + \frac{L_2(\sigma\lambda^H_{\max})^2}{2L_1^2\kappa^4}\|g_k\|^{2(2+p-2q)}. \tag{39}$$

Factoring out $\|g_k\|$:

$$\|g_{k+1}\| \le \|g_k\| \left(1 - \frac{\sigma\lambda_{\min}^H\gamma}{L_1\kappa^2}\|g_k\|^{1+p-2q} + \frac{L_2(\sigma\lambda_{\max}^H)^2}{2L_1^2\kappa^4}\|g_k\|^{2(2+p-2q)-1}\right).$$

Letting $u := \|g_k\|^{1+p-2q}$, we note that $2(2+p-2q)-1 = (1+p-2q)+(2+p-2q)$, so $\|g_k\|^{2(2+p-2q)-1} = u \cdot \|g_k\|^{2+p-2q}$. The contraction factor becomes:

$$\rho_k := 1 - u\left(\frac{\sigma\lambda_{\min}^H\gamma}{L_1\kappa^2} - \frac{L_2(\sigma\lambda_{\max}^H)^2}{2L_1^2\kappa^4}\|g_k\|^{2+p-2q}\right). \tag{40}$$

Step 4: Condition for gradient shrinkage. We need $\rho_k < 1$, i.e., the term in parentheses must be positive:

$$\frac{\sigma\lambda_{\min}^H\gamma}{L_1\kappa^2} > \frac{L_2(\sigma\lambda_{\max}^H)^2}{2L_1^2\kappa^4}\|g_k\|^{2+p-2q}.$$

Rearranging:

$$\|g_k\|^{2+p-2q} < \frac{2\sigma\lambda_{\min}^H\gamma L_1}{\kappa^2 L_2(\lambda_{\max}^H)^2}.$$

This must hold for all $k \ge k_0$ with $x_k \in B(x^\star, 2\delta)$. At entry, $\|g_{k_0}\| \le L_1\delta$, and since we will show gradients shrink, the worst case is at entry. So it suffices to require:

$$(L_1\delta)^{2+p-2q} < \frac{2\sigma\lambda_{\min}^H\gamma L_1}{\kappa^2 L_2(\lambda_{\max}^H)^2} \iff \delta < \frac{1}{L_1}\left(\frac{2\sigma\lambda_{\min}^H\gamma L_1}{\kappa^2 L_2(\lambda_{\max}^H)^2}\right)^{\frac{1}{2+p-2q}}. \tag{41}$$

Step 5: Effective contraction rate. Under condition (41), $\rho_k < 1$ for all $k \ge k_0$. Moreover, since $\|g_{k+1}\| \le \rho_k\|g_k\| < \|g_k\|$, the contraction rate ρ_k *improves* at each step (since smaller $\|g_k\|$ makes the negative terms in (40) smaller, but the positive term 1 stays). Therefore, the worst-case rate is at the first iteration:

$$\bar\rho := 1 - (L_1\delta)^{1+p-2q}\left(\frac{\sigma\lambda_{\min}^H\gamma}{L_1\kappa^2} - \frac{L_2(\sigma\lambda_{\max}^H)^2}{2L_1^2\kappa^4}(L_1\delta)^{2+p-2q}\right) < 1. \tag{42}$$

By induction, for all $m \ge 0$:

$$\|g_{k_0+m}\| \le \bar\rho^m\|g_{k_0}\| \le \bar\rho^m L_1\delta. \tag{43}$$

Step 6: Step length after m iterations. Using $\|d_k\| = \|H_k g_k\| \le \lambda_{\max}^H\|g_k\|$ and $\alpha_k = \frac{\sigma}{L_1\kappa^2}\|g_k\|^{1+p-2q}$:

$$\|x_{k_0+m+1} - x_{k_0+m}\| = \alpha_k\|d_k\| \le \frac{\sigma\lambda_{\max}^H}{L_1\kappa^2}\|g_{k_0+m}\|^{2+p-2q}.$$

Substituting (43):

$$\|x_{k_0+m+1} - x_{k_0+m}\| \le \frac{\sigma\lambda_{\max}^H}{L_1\kappa^2}(\bar\rho^m L_1\delta)^{2+p-2q} = \frac{\sigma\lambda_{\max}^H(L_1\delta)^{2+p-2q}}{L_1\kappa^2}\bar\rho^{m(2+p-2q)}. \tag{44}$$

Since $2+p-2q > 1$, this decays as $\bar\rho^{m(2+p-2q)}$, which is faster than $\bar\rho^m$.

Step 7: Distance to $x^\star$ after m iterations. From strong monotonicity ($\|g_k\| \ge \gamma\|x_k - x^\star\|$):

$$\|x_{k_0+m} - x^\star\| \le \frac{\|g_{k_0+m}\|}{\gamma} \le \frac{\bar\rho^m L_1\delta}{\gamma}. \tag{45}$$

Step 8: Triangle inequality. Combining Steps 6 and 7:

$$\begin{aligned}\|x_{k_0+m+1} - x^\star\| &\le \|x_{k_0+m+1} - x_{k_0+m}\| + \|x_{k_0+m} - x^\star\| \\ &\le \frac{\sigma\lambda^H_{\max}(L_1\delta)^{2+p-2q}}{L_1\kappa^2}\bar{\rho}^{m(2+p-2q)} + \frac{\bar{\rho}^m L_1\delta}{\gamma}.\end{aligned}$$

Since $2+p-2q > 1$, for any $m \ge 1$ we have $\bar{\rho}^{m(2+p-2q)} < \bar{\rho}^m$, so the first term is dominated by the second. Bounding:

$$\|x_{k_0+m+1} - x^\star\| \le \frac{\bar{\rho}^m L_1\delta}{\gamma}\left(1 + \frac{\sigma\lambda^H_{\max}\gamma(L_1\delta)^{1+p-2q}}{L_1\kappa^2}\right).$$

Define:

$$A := \frac{L_1}{\gamma}\left(1 + \frac{\sigma\lambda^H_{\max}\gamma}{L_1\kappa^2}(L_1\delta)^{1+p-2q}\right). \tag{46}$$

Step 9: Solve for m. We need $\|x_{k_0+m+1} - x^\star\| \le \delta$, i.e.:

$$\bar{\rho}^m \cdot A \le 1.$$

Taking logarithms. Since $\bar{\rho} < 1$, $\log\bar{\rho} < 0$, dividing flips the inequality:

$$m \ge \frac{\log A}{-\log\bar{\rho}}.$$

To bound $-\log\bar{\rho}$, write $\bar{\rho} = 1 - X$ where:

$$X := (L_1\delta)^{1+p-2q}\left(\frac{\sigma\lambda^H_{\min}\gamma}{L_1\kappa^2} - \frac{L_2(\sigma\lambda^H_{\max})^2}{2L_1^2\kappa^4}(L_1\delta)^{2+p-2q}\right) > 0.$$

Using $-\log(1-X) \ge X$ for $X \in (0,1)$:

$$\boxed{m^\star_{\mathcal{N}} \le \left\lceil \frac{\log A}{(L_1\delta)^{1+p-2q}\left(\frac{\sigma\lambda^H_{\min}\gamma}{L_1\kappa^2} - \frac{L_2(\sigma\lambda^H_{\max})^2}{2L_1^2\kappa^4}(L_1\delta)^{2+p-2q}\right)}\right\rceil} \tag{47}$$

where A is defined in (46), under the condition (41) on δ.

Verification: recovery of the restarted case. For $H_k = I$ ($\lambda^H_{\min} = \lambda^H_{\max} = 1$) and $p = q = 1$ ($1+p-2q = 0$, $2+p-2q = 1$):

- $(L_1\delta)^{1+p-2q} = 1$ and $(L_1\delta)^{2+p-2q} = L_1\delta$.
- The denominator becomes $\frac{\sigma\gamma}{L_1\kappa^2} - \frac{L_2\sigma^2}{2L_1^2\kappa^4}L_1\delta$. For $\sigma = \kappa = 1$: $\frac{\gamma}{L_1} - \frac{L_2\delta}{2L_1} = \frac{2\gamma - L_2\delta}{2L_1}$.
- $A = \frac{L_1}{\gamma}(1 + \frac{\sigma\gamma}{L_1\kappa^2}) \approx \frac{L_1}{\gamma}$ for typical parameters, so $\log A \approx \log(L_1/\gamma)$.
- This gives $m^\star \approx \frac{2L_1\log(L_1/\gamma)}{2\gamma - L_2\delta}$, matching $m^\star_{\mathcal{R}}$ from (37).

Remark 6. *The eigenvalues $\lambda^H_{\min}$ and $\lambda^H_{\max}$ of the implicit preconditioner H_k are not known in general for nonlinear conjugate gradient. Bounding them requires assumptions on the conjugate direction parameter β_k and is specific to each CG variant (FR, PRP+, HZ, etc.). For a specific β_k formula, these eigenvalues can be estimated, yielding a concrete expression for $m^\star_{\mathcal{N}}$.*

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